ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

MATHEMATICAL LITERACY P1

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 10 Mathematical Literacy Paper 1 practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark (M = method, A = accuracy). Give follow-through marks (CA) where a learner correctly uses an earlier wrong answer. Penalise once for missing units or the R sign.

QUESTION 1

1.1 $2\,500 + 850 + 1\,400 + 299 + 500 + 450 + 380$ ✓ $= \text{R}6\,379$ ✓ (2)

1.2 $6\,800 - 6\,379$ ✓ $= \text{R}421$ ✓ (2)

1.3 $\frac{2\,500}{6\,800} \times 100$ ✓✓ $= 36{,}76\%$ ✓ (3)

1.4 $2\,500 \times 8\% = \text{R}200$ ✓; $2\,500 + 200$ ✓ $= \text{R}2\,700$ ✓ (or $2\,500 \times 1{,}08$) (3)

1.5 $421 - 200 = \text{R}221$ ✓; yes, she still has R221 left ✓ (2)

1.6 $158 \times \text{R}2{,}85$ ✓ $= \text{R}450{,}30$ ✓ (2)

1.7 Price without VAT $= 345 \div 1{,}15$ ✓ $= \text{R}300$ ✓; VAT $= 345 - 300 = \text{R}45$ ✓ (3)

1.8.1 149×24 ✓ $= \text{R}3\,576$ ✓ (2)

1.8.2 $80 \times 24 = \text{R}1\,920$ ✓; $1\,999 + 1\,920$ ✓ $= \text{R}3\,919$ ✓ (3)

1.8.3 Option A ✓, by $3\,919 - 3\,576 = \text{R}343$ ✓ (2)

1.9 Any ONE sensible reason, e.g. for emergencies such as illness or job loss ✓✓ / to buy something big later without debt / to earn interest. (2)

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QUESTION 2

2.1 2.1.1: 2 500 m ✓ 2.1.2: 0,75 ℓ ✓ 2.1.3: $(3 \times 60) + 45 = 225$ min ✓ 2.1.4: 1,45 kg ✓ (4)

2.2.1 $8 \times 6{,}5$ ✓ $= 52\ \text{m}^2$ ✓ (2)

2.2.2 Area of one tile $= 0{,}5 \times 0{,}5 = 0{,}25\ \text{m}^2$ ✓; $52 \div 0{,}25$ ✓ $= 208$ tiles ✓ (3)

2.2.3 $208 \div 10 = 20{,}8$ ✓ $\rightarrow 21$ boxes (round up) ✓; $21 \times \text{R}189$ ✓ $= \text{R}3\,969$ ✓ (4)

2.2.4 $2 \times (8 + 6{,}5)$ ✓ $= 29$ m ✓ (2)

2.2.5 $29 \times \text{R}35$ ✓ $= \text{R}1\,015$ ✓ (2)

2.3.1 $3{,}142 \times 0{,}7^2 \times 1{,}5$ ✓✓ $= 2{,}31\ \text{m}^3$ ✓ (3)

2.3.2 $2{,}30937 \times 1\,000$ ✓ $= 2\,309{,}37\ \ell$ (accept $2\,310\ \ell$) ✓ (2)

2.4 $06{:}45 + 2$ h $= 08{:}45$ ✓; $+ 50$ min $= 09{:}35$ ✓ (2)

2.5 $(1{,}8 \times 25) + 32$ ✓ $= 77\ ^\circ\text{F}$ ✓ (2)

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QUESTION 3

3.1 1 unit (cm) on the map ✓ represents 50 000 of the same units (cm) in reality ✓. (2)

3.2 $4{,}2 \times 50\,000 = 210\,000$ cm ✓; $\div\ 100\,000$ ✓ $= 2{,}1$ km ✓ (3)

3.3 $2{,}1 \div 5 = 0{,}42$ h ✓; $0{,}42 \times 60$ ✓ $= 25{,}2$ min (about 25 minutes) ✓ (3)

3.4 D4 ✓ (1)

3.5 The shop ✓ (1)

3.6 South-east ✓✓ (2)

3.7 $3{,}5$ cm $\times\ 100 = 350$ cm $= 3{,}5$ m ✓; 4 cm $\times\ 100 = 4$ m ✓; area $= 3{,}5 \times 4$ ✓ $= 14$ m^2 ✓ (4)

3.8 Yes ✓. 1 : 50 000 is a larger scale: it shows a smaller area, so more detail can be shown ✓. (2)

3.9 So that real distances can be worked out from the map ✓✓ (accept: to show how much the area was made smaller). (2)

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QUESTION 4

4.1 19 ; 23 ; 27 ; 28 ; 30 ; 35 ; 35 ; 35 ; 41 ; 44 ; 47 ✓✓ (2)

4.2 35 ✓ (1)

4.3 The 6th value (middle of 11) ✓ $= 35$ ✓ (2)

4.4 Total $= 364$ ✓; $364 \div 11$ ✓ $= 33{,}09$ ✓ (3)

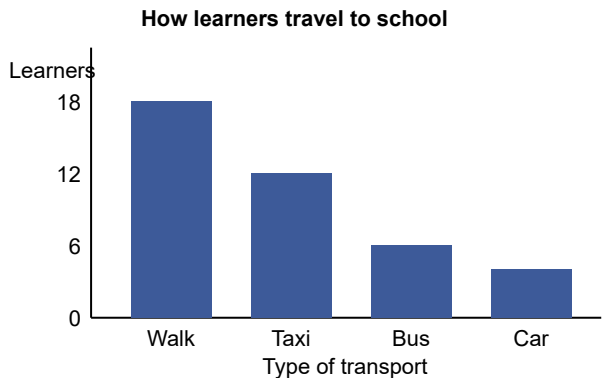
4.5 $47 - 19$ ✓ $= 28$ ✓ (2)

4.6 50% of $50 = 25$ marks ✓; learners below 25: 19 and 23, so 2 failed ✓ (5)

9 passed ✓; $\frac{9}{11} \times 100$ ✓ $= 81{,}82\%$ ✓

4.7 Median ✓. It is not pulled down by the low mark of 19 (outlier); it is the middle mark ✓. (Accept "mean" with a sensible reason, e.g. it uses all the marks.) (2)

4.8.1 (5)



Marks: heading ✓; both axes labelled ✓; even scale ✓; correct bar heights (all four) ✓✓

4.8.2 $\frac{18}{40} \times 100 \checkmark = 45\% \checkmark$ (2)

4.8.3 $\frac{12}{40} \checkmark = 0{,}3 \checkmark$ (2)

4.9 Discrete data can only take certain values (counted), e.g. number of learners ✓. (2)
Continuous data can take any value in a range (measured), e.g. height or mass ✓.

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TOTAL: 100

END OF MARKING GUIDELINE